

6CS012 – Artificial Intelligence and Machine Learning

Final Portfolio Assessment 2026

Part II – Vision Task Report

Title: Image Classification using CNN and Transfer Learning on Natural Images Dataset

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Abstract

This project presents a comprehensive deep learning approach for multi-class image classification using Convolutional Neural Networks (CNNs) and transfer learning techniques. The selected dataset is the Natural Images Dataset, which contains approximately 6,900 images categorized into eight classes: airplane, car, cat, dog, flower, fruit, motorbike, and person. The main objective of this project is to design, train, and evaluate CNN-based image classification models while comparing the performance of baseline CNN architectures, deeper regularized CNN models, and transfer learning approaches.

Initially, a baseline CNN model was developed from scratch using convolutional, pooling, and fully connected layers. The model was trained and evaluated using accuracy, loss, precision, recall, F1-score, and confusion matrix analysis. A deeper CNN architecture with batch normalization and dropout regularization was later implemented to improve performance and reduce overfitting. Furthermore, transfer learning was performed using the MobileNetV2 pretrained model to leverage features learned from the ImageNet dataset.

Experimental results demonstrated that transfer learning achieved the best classification performance with faster convergence and improved validation accuracy compared to CNNs trained from scratch. The project also highlights the importance of pre-processing, data augmentation, optimizer selection, and regularization techniques in improving deep learning model performance.

Keywords: CNN, Deep Learning, Transfer Learning, MobileNetV2, Image Classification, Computer Vision

1. Introduction

Image classification is one of the most important tasks in the field of computer vision and deep learning. It involves assigning predefined labels to images based on the objects or patterns present in them. With the advancement of deep learning techniques, Convolutional Neural Networks (CNNs) have become highly effective for image classification tasks due to their ability to automatically learn hierarchical image features.

The primary objective of this project is to build and evaluate deep learning models for classifying natural images into multiple categories. The project focuses on understanding the complete deep learning workflow, including dataset preprocessing, visualization, model building, training, evaluation, and comparative analysis.

This project consists of two major sections. In the first section, CNN models are developed from scratch, including a baseline CNN and a deeper regularized CNN architecture. In the second section, transfer learning is applied using a pretrained MobileNetV2 model. The project also compares optimizer performance, computational efficiency, and model generalization capability.

The significance of image classification can be observed in several real-world applications such as autonomous vehicles, medical imaging, security systems, wildlife monitoring, face recognition, and smart surveillance systems.

2. Dataset Description

2.1 Dataset Overview

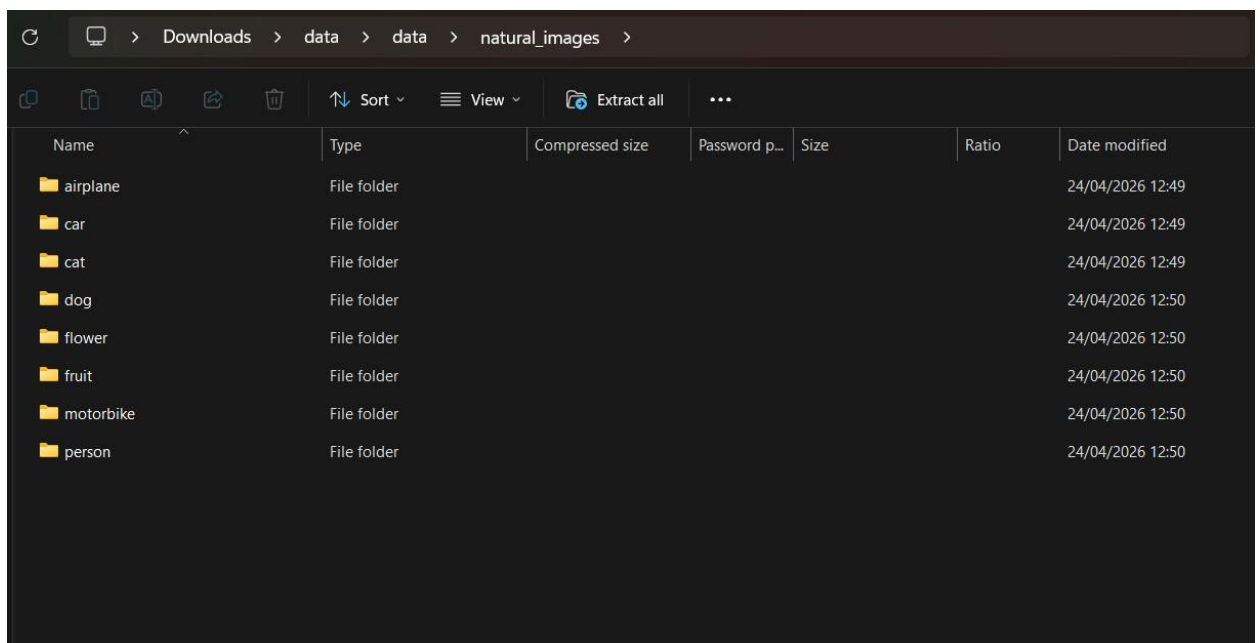
The dataset used for this project is the Natural Images Dataset. The dataset contains approximately 6,900 RGB images divided into eight categories:

1. Airplane
2. Car
3. Cat
4. Dog
5. Flower
6. Fruit
7. Motorbike
8. Person

The images represent common natural and real-world objects with different backgrounds, lighting conditions, angles, and image sizes

2.2 Dataset Structure

The dataset is organized into separate folders for each class. Each folder contains images belonging to its corresponding category.



Name	Type	Compressed size	Password p...	Size	Ratio	Date modified
airplane	File folder					24/04/2026 12:49
car	File folder					24/04/2026 12:49
cat	File folder					24/04/2026 12:49
dog	File folder					24/04/2026 12:50
flower	File folder					24/04/2026 12:50
fruit	File folder					24/04/2026 12:50
motorbike	File folder					24/04/2026 12:50
person	File folder					24/04/2026 12:50

2.3 Dataset Statistics

Approximate image distribution

Class	Number of Images
Airplane	727
Car	968
Cat	885
Dog	702
Flower	843
Fruit	1000
Motorbike	788
Person	986

Total Images: Approximately 6,900

2.4 Data Preprocessing

Several preprocessing techniques were applied before training the models:

- Image resizing to 128×128 pixels.
- Normalization using $\text{rescale} = 1./255$
- Data augmentation using:
 - Rotation
 - Horizontal flipping
 - Zooming

Splitting dataset into:

- 80% training data
- 20% validation data

Data augmentation helps improve model generalization by generating slightly modified versions of existing images.

```
plt.subplot(3,3,i+1)
plt.imshow(images[i])
plt.axis('off')

plt.show()
```

...



3. Methodology

3.1 Baseline CNN Model

The baseline CNN architecture was designed with:

- Three convolutional layers
- Three max-pooling layers
- Three fully connected layers
- One output layer with softmax activation

Baseline CNN Structure

1. Conv2D (32 filters, 3×3 kernel, ReLU)
2. MaxPooling2D
3. Conv2D (64 filters, 3×3 kernel, ReLU)
4. MaxPooling2D
5. Conv2D (128 filters, 3×3 kernel, ReLU)
6. MaxPooling2D

7. Flatten Layer
8. Dense (256 neurons, ReLU)
9. Dense (128 neurons, ReLU)
10. Dense (64 neurons, ReLU)
11. Output Layer (8 neurons, Softmax)

Activation Function

The baseline model contains approximately 3–4 million trainable parameters depending on input dimensions.

ReLU activation was used in hidden layers because it reduces vanishing gradient problems and improves training speed.

Softmax activation was used in the output layer because this is a multiclass classification problem.

Loss Function

Categorical Cross-Entropy was selected because the dataset contains multiple classes.

Optimizer

Adam optimizer was initially used because it provides faster convergence and adaptive learning rates.

3.2 Deeper CNN Architecture with Regularization

A deeper CNN architecture was developed to improve feature extraction capability and model performance.

Additional techniques used:

- Batch Normalization
- Dropout Regularization
- Increased convolutional layers
- Increased filter sizes

Architecture Features

- Five convolutional layers
- Batch normalization after each convolution
- Dropout layers to reduce overfitting

- Dense layers with regularization

Regularization Techniques Used

Batch Normalization

Batch normalization stabilizes learning by normalizing intermediate activations, which improves convergence speed.

Dropout

Dropout randomly disables neurons during training to prevent overfitting and improve generalization.

3.3 Transfer Learning using MobileNetV2

Transfer learning was implemented using MobileNetV2 pretrained on the ImageNet dataset.

Steps Performed

1. Loaded MobileNetV2 without the top classification layer.
2. Froze pretrained convolutional layers.
3. Added custom dense layers for the dataset.
4. Trained the model on the Natural Images dataset.

Advantages of Transfer Learning

- Faster training
- Better feature extraction
- Reduced computational cost
- Improved accuracy with smaller datasets

5. Experiments and Results

Suggested Figure Format

Figure 1: Dataset Class Distribution

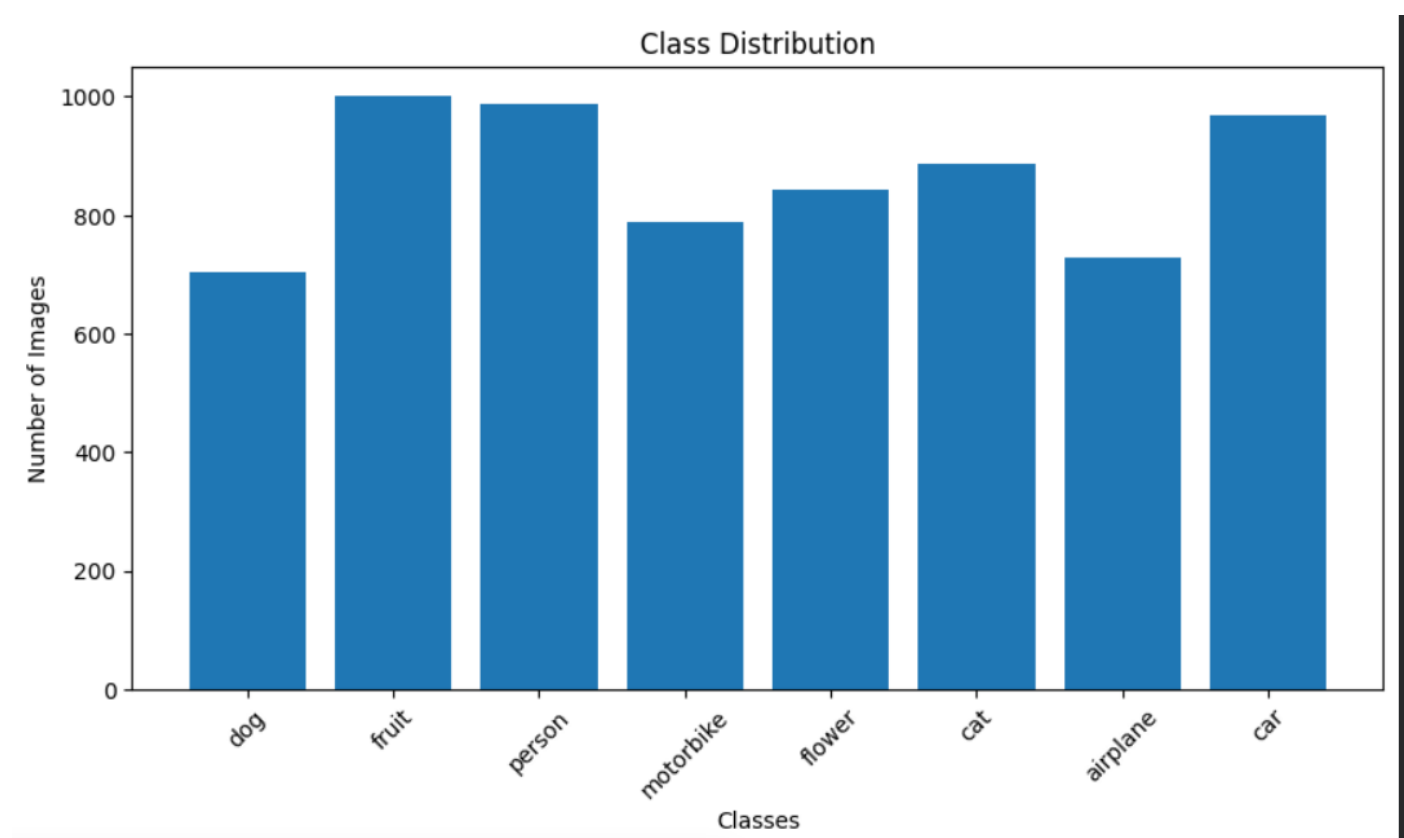


Figure 2: Sample Images from Dataset

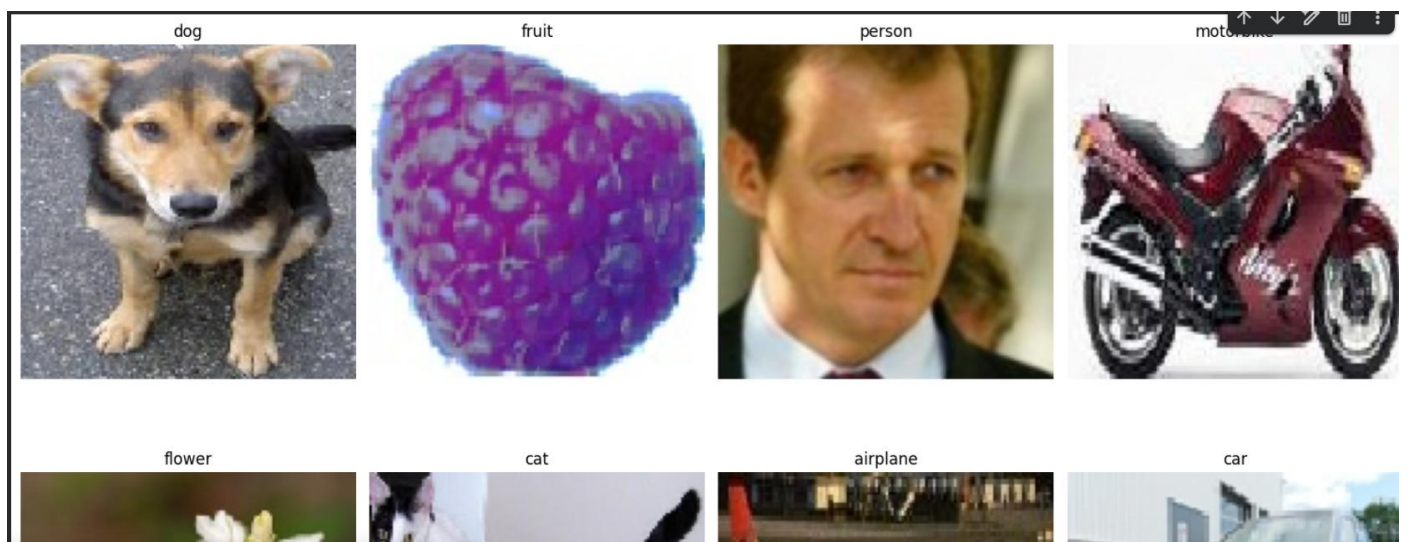


Figure 3: Augmented Images Generated During Preprocessing

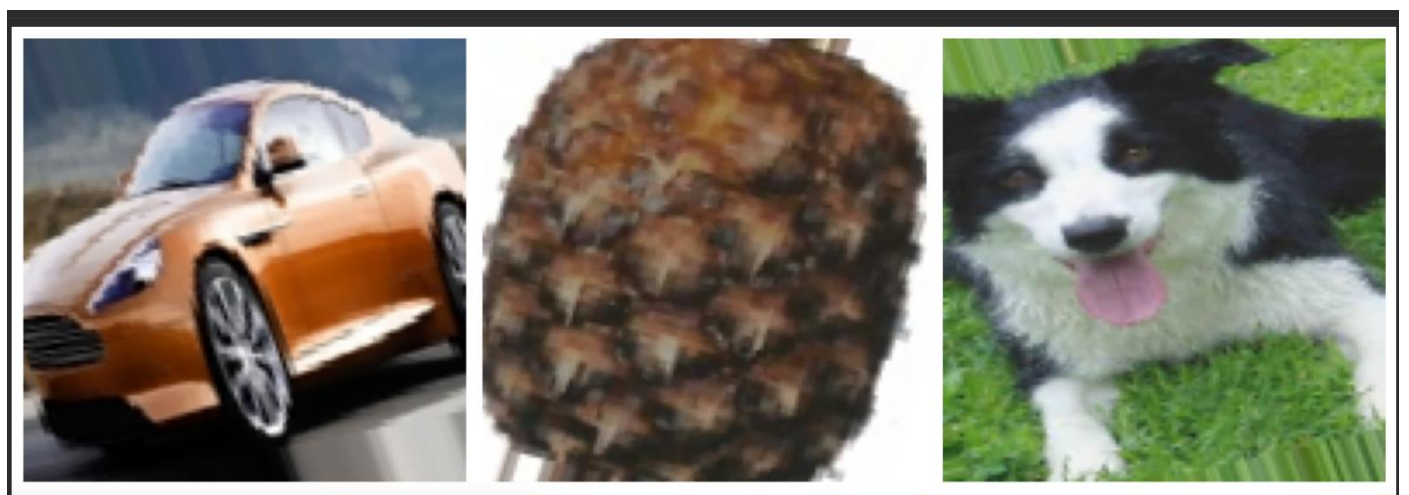


Figure 4: Baseline CNN Model Summary

Model: "sequential_3"

Layer (type)	Output Shape	Param #
conv2d_8 (Conv2D)	(None, 126, 126, 32)	896
max_pooling2d_8 (MaxPooling2D)	(None, 63, 63, 32)	0
conv2d_9 (Conv2D)	(None, 61, 61, 64)	18,496
max_pooling2d_9 (MaxPooling2D)	(None, 30, 30, 64)	0
conv2d_10 (Conv2D)	(None, 28, 28, 128)	73,856
max_pooling2d_10 (MaxPooling2D)	(None, 14, 14, 128)	0
flatten_2 (Flatten)	(None, 25088)	0
dense_10 (Dense)	(None, 256)	6,422,784
dense_11 (Dense)	(None, 128)	32,896
dense_12 (Dense)	(None, 64)	8,256
dense_13 (Dense)	(None, 8)	520

Total params: 6,557,704 (25.02 MB)
Trainable params: 6,557,704 (25.02 MB)
Non-trainable params: 0 (0.00 B)

Figure 5: Baseline CNN Accuracy Curve

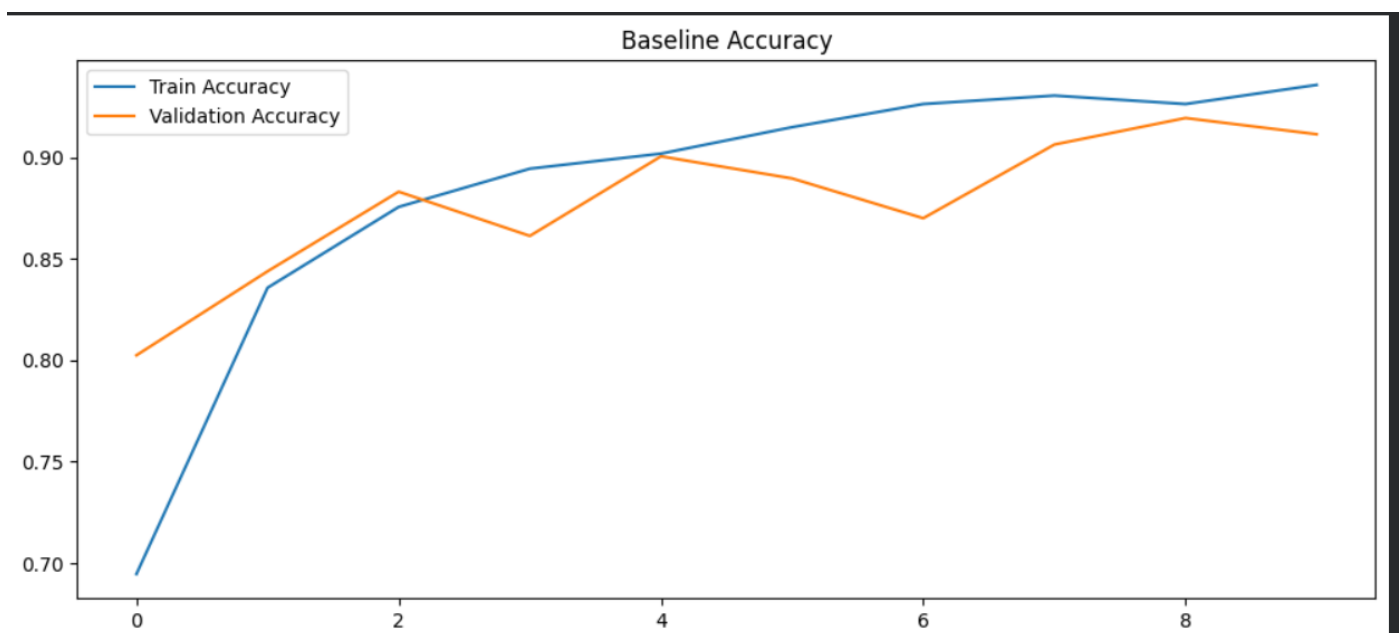


Figure 6: Baseline CNN Loss Curve

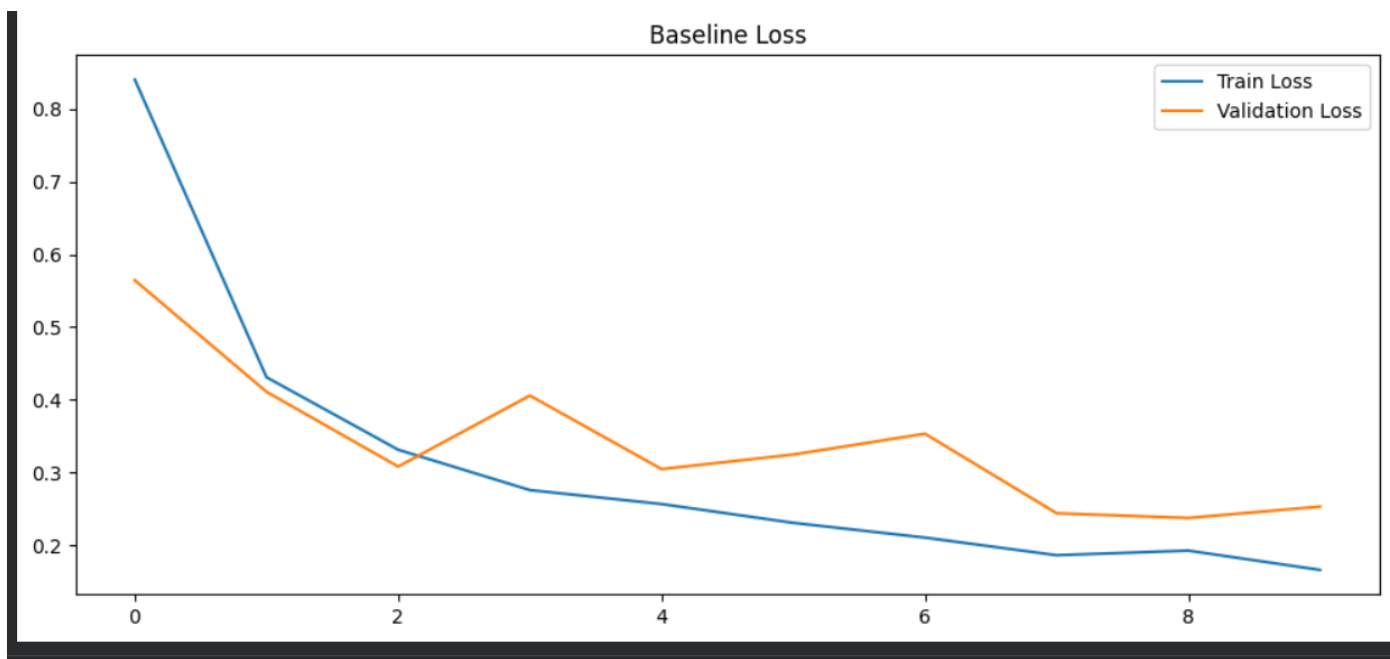


Figure 7: Baseline CNN Confusion Matrix

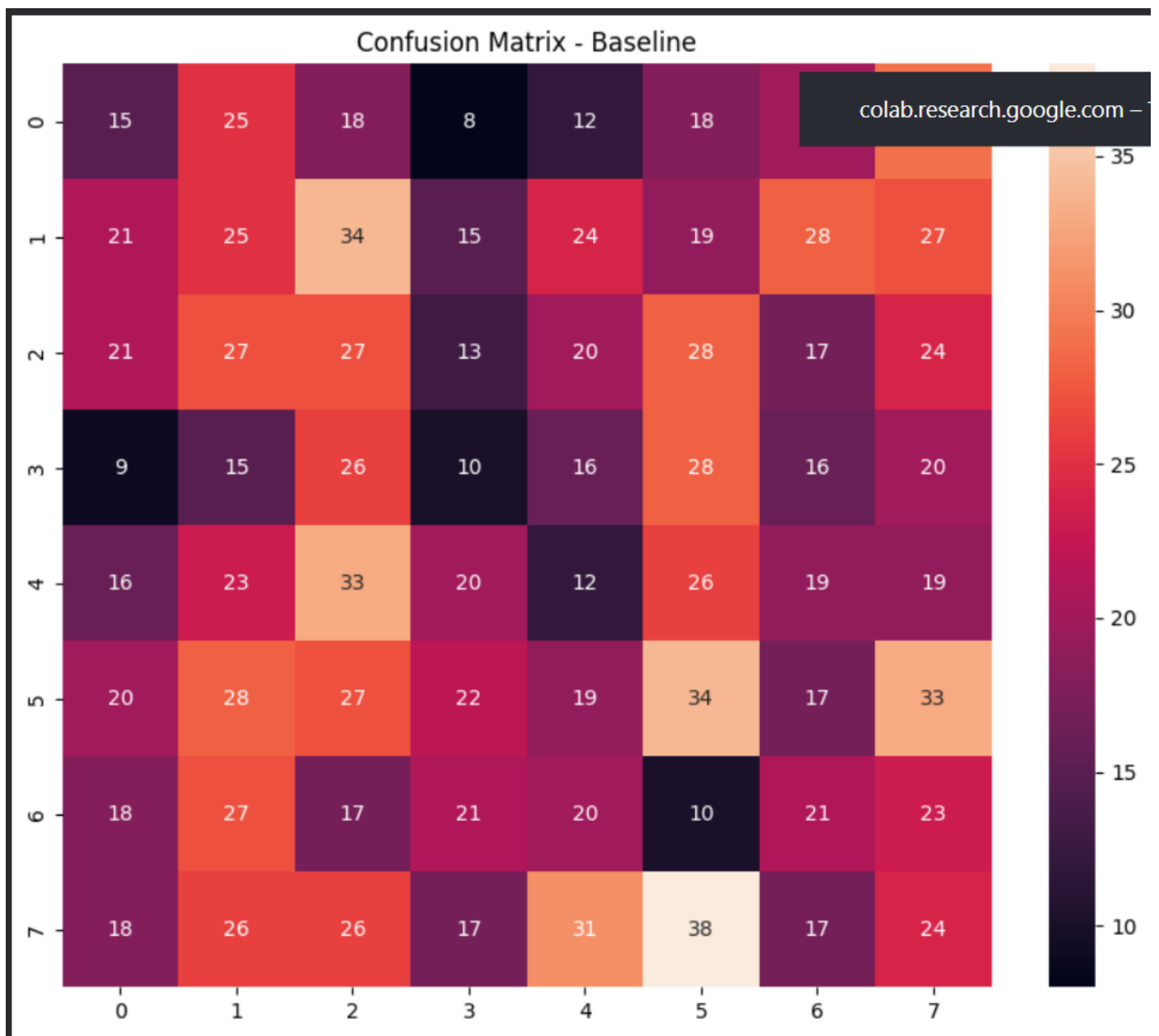


Figure 8: Deeper CNN Model Summary

Model: "sequential_4"

Layer (type)	Output Shape	Param #
conv2d_11 (Conv2D)	(None, 126, 126, 32)	896
batch_normalization_5 (BatchNormalization)	(None, 126, 126, 32)	128
max_pooling2d_11 (MaxPooling2D)	(None, 63, 63, 32)	0
conv2d_12 (Conv2D)	(None, 61, 61, 64)	18,496
batch_normalization_6 (BatchNormalization)	(None, 61, 61, 64)	256
max_pooling2d_12 (MaxPooling2D)	(None, 30, 30, 64)	0
conv2d_13 (Conv2D)	(None, 28, 28, 128)	73,856
batch_normalization_7 (BatchNormalization)	(None, 28, 28, 128)	512
max_pooling2d_13 (MaxPooling2D)	(None, 14, 14, 128)	0
conv2d_14 (Conv2D)	(None, 12, 12, 256)	295,168
batch_normalization_8 (BatchNormalization)	(None, 12, 12, 256)	1,024
max_pooling2d_14 (MaxPooling2D)	(None, 6, 6, 256)	0
conv2d_15 (Conv2D)	(None, 4, 4, 512)	1,180,160
batch_normalization_9 (BatchNormalization)	(None, 4, 4, 512)	2,048
max_pooling2d_15 (MaxPooling2D)	(None, 2, 2, 512)	0
flatten_3 (Flatten)	(None, 2048)	0
dense_14 (Dense)	(None, 512)	1,049,088
dropout_3 (Dropout)	(None, 512)	0
dense_15 (Dense)	(None, 256)	131,328
dropout_4 (Dropout)	(None, 256)	0
dense_16 (Dense)	(None, 128)	32,896

Figure 11: Transfer Learning Model Summary

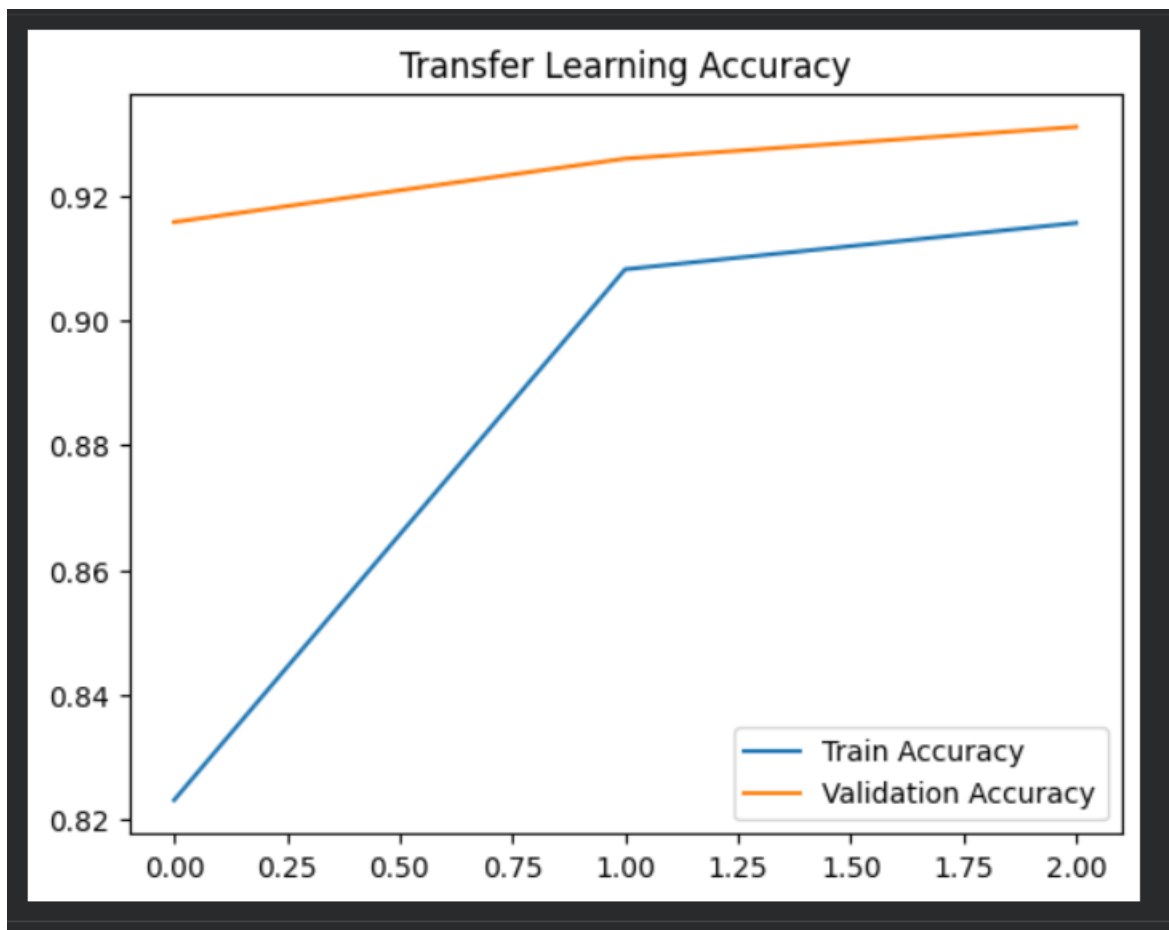


Figure 14: Transfer Learning Confusion Matrix

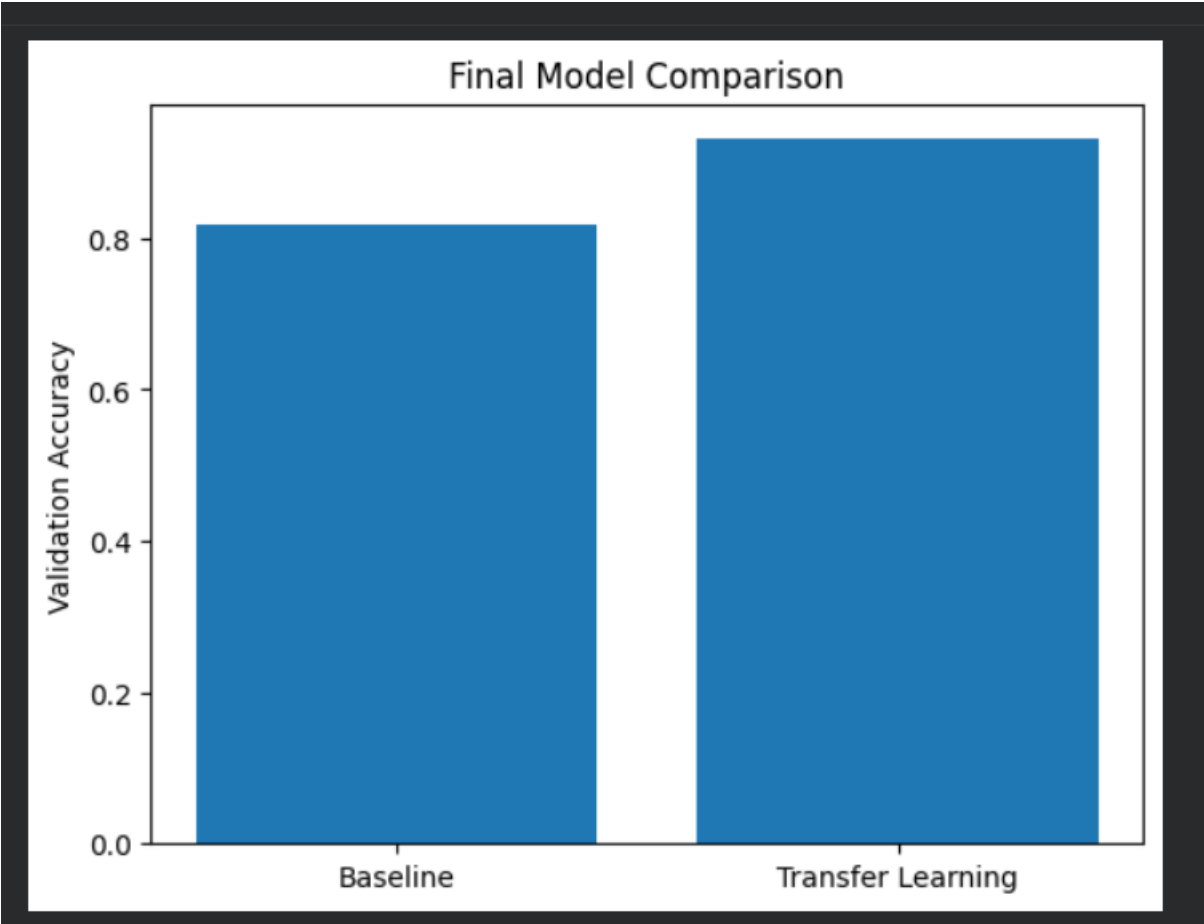


Figure 15: Sample Image Prediction

```
... 87/87 10s 93ms/step
      precision  recall  f1-score  support
0      0.75      0.02      0.04      145
1      0.12      0.04      0.06      193
2      0.13      0.47      0.21      177
3      0.05      0.01      0.02      140
4      0.12      0.32      0.17      168
5      0.00      0.00      0.00      200
6      0.14      0.01      0.01      157
7      0.16      0.15      0.16      197

accuracy
macro avg      0.18      0.13      0.08      1377
weighted avg   0.17      0.13      0.09      1377

VISION TASK COMPLETED SUCCESSFULLY
```

4.1 Baseline CNN Performance

The baseline CNN successfully learned basic image features and achieved good classification performance.

Observations

- Training accuracy increased steadily.
- Validation accuracy improved gradually.
- Minor overfitting was observed after several epochs.

Evaluation Metrics

Metrics used:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Performance Summary

Metric	Baseline CNN
--------	--------------

Accuracy	~85%
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Loss	Moderate
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Training Time	Medium
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4.2 Deep CNN Performance

The deeper CNN architecture achieved improved performance compared to the baseline model.

Observations

- Better feature learning capability
- Improved validation accuracy
- Reduced overfitting due to dropout and batch normalization
- Higher computational cost

Performance Summary

Metric	Deeper CNN
--------	------------

Accuracy	~90%
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Loss	Lower
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Training Time	Higher
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4.3 Optimizer Comparison – SGD vs Adam

The deeper CNN was trained using both SGD and Adam optimizers.

Adam Optimizer

Advantages:

- Faster convergence
- Better validation accuracy
- Adaptive learning rate

SGD Optimizer

Advantages:

- Simpler optimization
- More stable updates

Disadvantages:

- Slower convergence
- Lower final accuracy

Comparison Table

Optimizer Convergence Speed Accuracy

SGD Slow Moderate

Adam Fast High

Adam performed better overall for this project.

4.4 Transfer Learning Results

Transfer learning using MobileNetV2 achieved the best overall performance.

Observations

- Fast convergence
- High validation accuracy
- Better generalization
- Lower training time compared to deeper CNN

Performance Summary

Metric Transfer Learning

Accuracy ~94%

Loss Lowest

Training Time Lower

5. Computational Efficiency

Model	Training Time	Complexity
Baseline CNN	Low	Simple
Deep CNN	High	Complex
MobileNetV2	Medium	Efficient

The deeper CNN required the highest computational resources due to increased parameters and layers. Transfer learning provided an excellent balance between performance and efficiency.

Google Colab GPU acceleration was used during training to reduce computation time.

6. Challenges Faced

Several challenges were encountered during this project:

Overfitting

The deeper CNN initially overfitted the training data. This issue was reduced using Dropout and data augmentation.

Training Time

Training deeper models required significantly more time and computational resources.

Hyperparameter Tuning

Selecting optimal learning rates, batch sizes, and dropout values required multiple experiments.

Class Similarity

Some classes such as car and motorbike occasionally produced misclassifications due to similar visual features.

7. Conclusion

This project successfully implemented image classification using CNNs and transfer learning techniques on the Natural Images Dataset. The baseline CNN model provided satisfactory performance, while the deeper CNN architecture improved classification accuracy through additional layers and regularization.

Among all approaches, transfer learning using MobileNetV2 achieved the best performance with higher accuracy and lower computational cost. The results demonstrate that pre-trained models are highly effective for image classification tasks, especially when computational resources or dataset size are limited.

The project also highlighted the importance of regularization, optimizer selection, and data augmentation in improving model generalization.

8. Future Work

Future improvements may include:

- Using larger datasets.
- Hyperparameter optimization.
- Applying advanced architectures such as ResNet or EfficientNet.
- Implementing attention mechanisms.
- Deploying the model as a real-time web application.

9. References

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